APMP Historical Data Lake: Enabling Point-in-Time Analysis of Salesforce Operational Data

Version 1.1

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Purpose

Salesforce is a vital system for managing operational data across accounts and activities in the *All of Us* Program Management Platform (APMP). However, a key limitation is the inability to perform comprehensive "look back" analyses on participant account data as a whole. While some individual data elements have standard historical tracking in Salesforce out-of-the-box, there is a limited number of fields that can be tracked and the aggregate or composite data across all fields on accounts lacks a consistent temporal snapshot. This limitation impairs analytic capabilities, particularly for dashboards requiring dynamic date sliders to explore trends over time or at specific time points.

The APMP Historical Data Lake was developed to address this gap by capturing daily snapshots of key Salesforce objects in Production from June 25, 2026, moving forward. This enabled robust historical trend analysis and the capacity to analyze the state of account data at any given point in time.

The APMP Historical Data Lake provides a scalable, cost-effective foundation for historical Salesforce analytics by preserving daily point-in-time snapshots of operational data. It enables accurate trend analysis, supports advanced reporting and dashboarding capabilities, reduces long-term data retention costs, and empowers stakeholders to make more informed decisions based on a complete historical view of participant and operational activity.

Design

The APMP Historical Data Lake is built on a modern, cloud-native AWS architecture that transforms how the organization captures, stores, and interrogates Salesforce data over time. Its core design principles are full automation, cost efficiency, and analytical depth.

- **Nightly Automated Bulk Extraction:** At midnight UTC, a fully automated pipeline — running on AWS ECS Fargate — spins up an on-demand containerized task for each configured Salesforce object. The task authenticates to Salesforce using

credentials stored securely in AWS Secrets Manager and performs a full bulk export of the object's current state. Unlike Salesforce's native history tracking, which records only selected field changes and retains them for a limited window, this approach preserves the *entire* state of every record each night. The result is an authoritative, immutable audit trail that enables the organization to reconstruct exactly what the data looked like on any given date — past or present.

- **Two-Tier Cloud Data Lake Storage:** Raw nightly snapshots land in a dedicated AWS S3 raw bucket, encrypted at rest with AWS Key Management Service (KMS)-managed keys and protected against public access. A second processed bucket holds the curated analytics-ready data. Both tiers are designed to grow with the organization's needs without architectural changes and integrate directly with AWS-native analytics tooling. All S3 traffic is routed internally through a Virtual Private Cloud (VPC) gateway endpoint, eliminating data transfer costs.
- **Automated SCD2 Processing and Intelligent Partitioning:** At 2 AM UTC—after each night's extraction completes—an AWS Lambda function triggers an AWS Glue distributed ETL job, passing the prior day's date as a parameter. The Glue job applies Slowly Changing Dimension Type 2 (SCD2) logic, meaning every change to a record is preserved as a versioned row with effective-date ranges rather than overwriting prior values. The processed data is written in Apache Iceberg table format, organized by Salesforce object type, account identifier, and date. This partitioning strategy means that time-bound or account-scoped queries retrieve only the relevant data—dramatically improving query speed and reducing compute costs at scale.
- **Self-Service Analytics via Amazon Athena:** The Iceberg tables in the processed bucket are registered in the AWS Glue Data Catalog and are directly queryable through Amazon Athena using standard SQL—no infrastructure to provision, no data movement required. This gives analysts and reporting stakeholders point-in-time query capability: the ability to ask, "what did this account's data look like on a specific date?" and get an immediate, accurate answer.
- **Significant Cost Reduction Over Native Salesforce Storage:** Salesforce storage is among the most expensive line items in any organization's Salesforce contract, with costs scaling steeply as data volume grows. By offloading historical snapshots to AWS S3—where storage costs are orders of magnitude lower—the organization can retain years of history affordably, archive infrequently accessed datasets to cold storage tiers, and avoid the Salesforce storage overages that would otherwise accompany long-term historical retention.

Solution Outcomes

The APMP Historical Data Lake transforms the way Salesforce operational data is analyzed:

- **Point-in-Time Analytics:** Analysts can now reconstruct the full state of account data as of any historical date once in our production environment permitting true temporal comparisons.
- **Enhanced Dashboards:** Integration with Salesforce Data360, Salesforce CRMA, and Tableau Next analytic platforms allows dynamic date sliders and temporal filters to drive dashboards that reflect the precise data configuration for selected points in time.
- **Operational Insights:** Analysts can now better structure date sliders on dashboards so that the full picture of the dashboard shows historical metrics for more accurate operational analytics. Trends in operational enhancements and efficiencies can be monitored over time more effectively.
- **Scalability and Flexibility:** The AWS data lake architecture supports future scaling to additional Salesforce objects, more frequent snapshots, or integration with other data sources, such as Salesforce Data 360. Salesforce tools such as Tableau Next and Data 360, which utilize a credit model for queries, can offload some of the more complex historical queries to other tools.

The APMP Historical Data Lake thus provides a foundational capability for rich historical analysis that was not previously possible within Salesforce alone, leveraging AWS to deliver actionable intelligence on evolving operational data.

Salesforce Objects Supported

The APMP Historical Data Lake currently supports daily snapshots of the Salesforce Account, AccountContactRelation, and EventLogFile objects. Additional objects are anticipated to be added.

Data Schema

Each Salesforce object tracked by the data lake is represented as a single, continuously growing history table in the AWS Glue Data Catalog. All original Salesforce field names are preserved and normalized to lowercase for SQL compatibility, meaning analysts query

familiar fields such as id, name, type, annual_revenue, and lastmodifieddate exactly as they appear in Salesforce—without translation.

Every row in a history table represents a version of a record, not just its current state. The ETL process appends four metadata columns to each row to track its place in time:

- valid_from: The date this version of the record became active
- valid_to: The date this version was superseded (NULL if still current)
- is_current: Boolean flag for fast filtering of the present-day state
- loaded_at: Timestamp of when this row was written to the data lake

When a record changes in Salesforce, the prior version's valid_to is set to the change date and is_current is flipped to false. A new row is inserted with the updated field values and is_current = true. Records that have not changed since their last appearance are intentionally skipped to avoid inflating the table with duplicate rows.

Tables are physically partitioned by year and month of valid_from, meaning the storage layer groups data into date-based folder hierarchies on S3. This has a direct operational benefit: a query scoped to a specific time window—say, "show me all accounts as of Q3 2025"—scans only the relevant monthly partitions rather than the entire table, keeping query times fast and Athena compute costs low regardless of how many years of history accumulate.

Nightly snapshots of only updated records are downloaded at midnight-1am depending on the daylight savings time status. The data is then scheduled to be loaded into the tables 3 hours later to be available in the data lake.

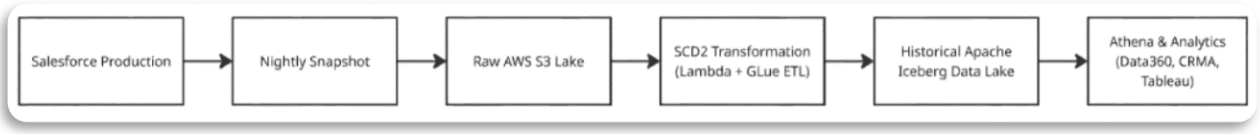


Figure 1: APMP Data Lake Data Processing Workflow

Version	Date	Modified By	Notes
1.0	6/12/2026	Lydia Luy Tan	Initial draft
1.1	6/26/2026	Lydia Luy Tan	Updated Production snapshot date. Added flow diagram